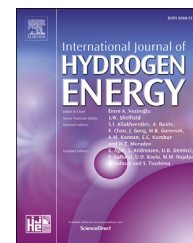


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Load profile based empirical model for the lifetime prediction of an automotive PEM fuel cell

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ABSTRACT

The prediction of the lifetime of a PEM fuel cell under a specific electrical load profile is beneficial for automotive applications. An empirical lifetime prediction model coupled with an electrical load profile is presented based on an estimate of the degradation rate of the cell voltage. First, a cause and effect chain-based analysis of the PEM fuel cell's degradation process is presented. Second, a characteristic value is extracted and is treated as a numerical measurement of the effect of load profile on voltage degradation. The characteristic value is defined using both load current and load voltage. Third, an empirical model is proposed, modeling the voltage degradation rate as a linear function of the load profile's characteristic value. Finally, experimental data are used to fit the coefficients and validate the model.

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Introduction

Polymer electrolyte membrane (PEM) fuel cell is one of the most promising energy converters for automobiles. Though significant progress has been achieved over the last two decades, the low durability of PEMFCs under a wide range of operating conditions still remains a critical technical barrier to their commercialization. The electrical load cycles, caused by the vehicle's driving cycles, are the most important factors hindering durability performance. When a PEMFC serves as the main power source, the load profile is closely determined by the driving cycles, and in hybrid power source system, the load profile is loosely related to the driving cycles through

system configuration and energy management strategies [1,2]. Because the load profile of PEMFCs is likely precalculated with numerical or simulation tools [3], e.g., Matlab/Simulink, AVL Cruise, etc. during the design of the automotive powertrain, the lifetime prediction model considering the load profile is beneficial to fuel cell design and optimization.

In previous research, specific degradation models have been developed at components or cells level, but few aimed at stack's synthetic performance with regards to different load profiles. For the perfluorinated sulfonic acid (PFSA) ionomers membrane, Chemical degradation dominates the perfluorinated sulfonic acid (PFSA) ionomers at early age degradation, while operation condition cycling accelerates later. Xie et al. [4] developed a kinetic model, which could quantitatively

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distinguish between weak polymer end groups versus side chain cleavage. Shim et al. [5] modeled membrane thinning behaviors with a simple first-order reaction with the effect of partial pressure taken into account. M. Gummall et al. [6] developed a physics-based theoretical model to predict the chemical degradation considering transport and reaction of crossover gases, hydrogen and oxygen. Wong et al. [7] developed a model for simulation of in-situ chemical degradation, which treated H_2O_2 formation via two-electron-transfer at the anode and radical formation from a single Fenton's reaction. In the following work [8], the model is expanded to incorporate the transport of Fe^{2+} and Fe^{3+} in the catalyst layers. Aindow et al. [9] adopted a stress vs. cycles-to-failure curve to predict durability of the membranes under humidity cycling, and assess the fraction of membrane life remaining in degraded MEAs. Burlatsky et al. [10] used three stress model for predicting the life of membranes concerning hydration cycling. Kusoglu et al. [11] presented a theoretical modeling framework for pinhole growth in the polymer membrane due to environmentally-driven cyclic loads. Quiroga et al. [12] proposed a multi-paradigm model on the membrane degradation, by combining coarse grained molecular dynamics and a multiscale cell performance model. Shah et al. [13] incorporated the membrane chemical degradation process to a PEM fuel cell model, unzipping of the polymer backbone and cleavage of the side chains. Platinum dissolution, coarsening, coalescence and carbon corrosion are the main mechanism of catalyst degradation. Darling et al. [14] initially proposed a model for platinum dissolution and movement in the MEA. Li et al. [15] developed an one-dimensional model to study platinum degradation and the subsequent electrochemical surface area. Ghelichi et al. [16] studied the chemical degradation model of fuel cell ionomer. They also simplified the set of model equations, and achieved an analytical expression focusing on the hydroxyl radicals as a function of initial concentrations of iron ions and hydrogen peroxide. Malek et al. [17] comprised a versatile multiscale model of carbon corrosion processes, which focused on water morphology and microstructure changes. Kulikovskiy et al. [18] reconstructed an equation with the local current, exchange current density and GDL oxygen diffusivity, concerning the non-uniform aging of the cathode catalyst along the channel. Robin et al. [19] combined a mechanistic catalyst dissolution model to a dynamic PEMFC cell model to predict the performance loss, and the equivalent active surface area was used as a key index. Wu&Fuller [20] proposed a similar model that includes also the formation of a Pt band in the membrane. Franco et al. [21] built a multiscale atomistic/kinetic model is derived providing mechanistic insights on the impact of the nanostructure and operating conditions on Pt alloy catalyst. Maleki et al. [22] proposed a neural network model for Pt/C cathode degradation. Gas diffusion layer degradation is mainly caused by the loss of hydrophobicity due to PTFE degradation. Seidenberger et al. [23] developed 3D Monte Carlo model in order to investigate the effect of the PTFE degradation on the water distribution.

Models aimed to predicate PEMFC cell or stack lifetime are practically useful for automotive application. Fowler et al. [22] first introduced an aging term for use in a generalized steady state model. In this model, two semi-empirical parameters

were applied, the equilibrium water content and the catalytic activity. Lee et al. [25] followed up on Fowler's idea, used four estimated parameters for global fuel cell voltage data, and considered the contributions of the loss of catalytic activity and membrane/electrode conductivity. Jomori et al. [26] developed a quantitative model to estimate the performance decay of membrane-electrode assembly due to the increase in oxygen transport resistance. Kim et al. [27] proposed a prognostic method for the health degradation estimation, which was based on equivalent circuit and impedance spectrum. Lu et al. [28] fitted a semi-empirical voltage degradation model, which assumed that the ohmic voltage loss and the leakage current functioned as a linear increment with regard to time. This model was validated under specific dynamic cycling. Pei et al. [29] proposed a purely empirical formula to calculate the lifetime of an automotive PEMFC, taking four factors into consideration: load cycle times, start–stop cycle times, idle time, and high power load time. In Pei's formula, four typical operating conditions had to be extracted from the loading profile. Placca et al. [30] performed a fault tree analysis of the fuel cell degradations process, and obtained a statistical model to predict degradation probability. Lafy et al. [31] studied the parameter variation during life of the stack based on empirical equation and equivalent electrical impedance model; Wu et al. [32] carried out a nonlinear self-adaptive model to predict the performance based on relevance vector machine. Jouin et al. [32] integrating some of the most recent understanding of the phenomena serves as a basis for a complete degradation and failure analysis.

In automotive applications, the loading profile is considered the main factor affecting the lifetime of the PEMFC. The objective of this study is to model the voltage degradation behavior under different load profiles and to enable the prediction of the lifetime of a PEMFC. A characteristic value is proposed to enable the measurements of different load profiles to perform comparisons for determining the effects on the degradation of the PEMFC. The characteristic value is defined for load current and load voltage and is multiplied by weight functions. Experimental data from the literature and from experiments are acquired to support the empirical model.

Characteristic value of load profile measurement for PEMFC degradation

A general review: cause and effect chain of automotive PEMFC degradation

The load profile of an automotive PEMFC is mainly influenced by driving cycles, such as speed cycles, acceleration, start/stop, idling, etc. The PEMFC's load profile experiences many features for different time segments [33–35]: 1) constant load current or galvanostatic operation during constant-speed cruising; 2) current cycling during speeding up or slowing down; 3) extra-low current or open-circuit voltage during idling/starting; 4) extra-high current during rapid acceleration.

The key factors affecting voltage degradation during the lifetime of a PFSA ionomer membrane based electrolyte and a Pt/C-based catalyst PEMFC are shown in Fig. 1.

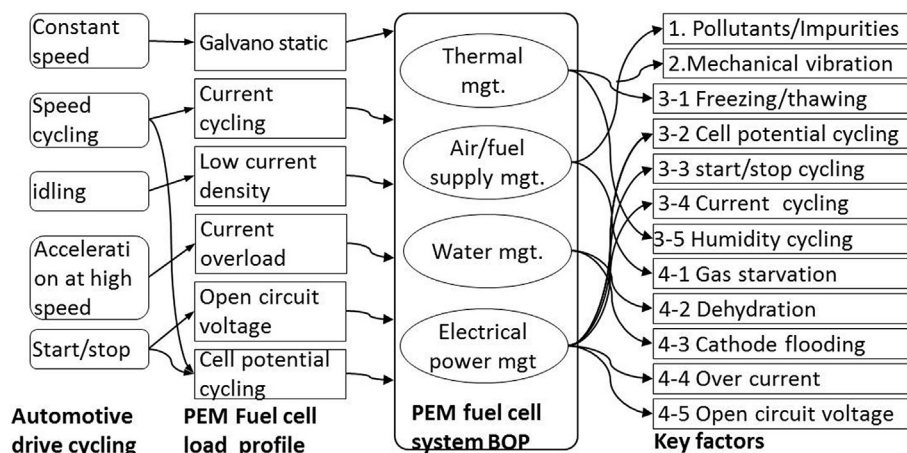


Fig. 1 – Factors affecting voltage degradation in an automotive PEMFC.

Automotive driving cycles influence the PEMFC's load profile, and the load profiles result in key factors that affect voltage degradation. Fig. 1 shows the following: 1) pollutants/impurities usually originate from unsuitable hydrogen fuel or contaminated air; 2) the mechanical vibration is affected partly by the road surfaces [36], and partly by the differential pressures of the anode/cathode gas; 3) key factors frequently encountered during normal operation are freezing/thawing, potential cycling, start/stop cycling, current cycling, and humidity & thermal cycling (noted as factors 3-1/2/3/4/5); 4) key factors during malfunctions (noted as factors 4-1/2/3/4/5) are gas starvation, membrane dehydration, anode/cathode flooding, overcurrent, and open circuit voltage. The factors during normal and malfunction operations are caused either directly or indirectly by the PEMFC's load profile.

A cause and effect chain is presented in Fig. 2, focusing on PFSA membrane, Pt/C catalyst and GDL which are the key components limiting stack lifetime [37,38]. Typical phenomena and fault modes are connected by $\rightarrow$ as cause-effect chain, and the degradation process between begin-of-life (BOL) and end-of-life (EOL) is generally divided into two phases: early age and late age.

In Fig. 2, three kinds of degradation mechanism are defined: direct intensification, positive feedback and cross cancellation:

- 1) Direct intensification mechanism. Factors that clearly accelerate the degradation process are marked firstly with $\square \rightarrow$, and the notation numbers in the box represent specific factors from Fig. 1. The key factors aggravate the degradation process through different mechanisms. For example, membrane dehydration (4-2 in Fig. 2) accelerates a membrane's early-age corrosion, while freezing/thawing, start/stop, and potential/current cycling (3-1/2/3/4 in Fig. 2) affect the membrane's corrosion in the long term. Open circuit voltage/high potential (4-5 in Fig. 2) result in carbon corrosion and Pt particles dissolve, redeposit, agglomerate, and redistribute at an early age. Overcurrent (4-4 in Fig. 2) becomes very harmful after pinholes appear.
- 2) Interactive intensification mechanism. Some factors are not caused from operation condition but from adjacent component. For example, the lost Pt particle from catalyst

layer and carbon from GDL can be cation pollution to membrane; at the same time, penetrated oxygen helps to carbon support corrosion and Pt migration.

- 3) Positive feedback mechanism. Positive feedback is a process (used frequently in control theory), that occurs in a feedback loop in which the effects of a small disturbance on a system include an increase in the magnitude of the perturbation. For example, gas penetration through membrane induces exothermic direct combustion which likely causes local hot spot, and the hot spot enlarges the pin hole which helps to the gas penetration. Both the positive feedback and interactive intensification are marked with $\rightarrow$.
- 4) Synergic cancellation mechanism. Under certain cases, the degradation effect simultaneously under two or more factors can't be added, but partly canceled. For example, Franco et al. [39] found that the long-term degradation of the PEFC performance under the current cycling is lower under $H_2 + CO$ compared to pure H_2 operation, which mean CO plays cancellation effects on current cycling in certain cases. The synergic cancellation mechanism is marked with $\circ \rightarrow$.

In the early age of degradation, as shown in Fig. 2, direct intensification mechanism plays a leading role and the components degrade gradually to a certain degree. Chemical damages and the decomposition of the sulfonated acid group result in thinning of the membrane [40,41]. Thermal and electrochemical mechanisms dominate the process while the initial mechanical defects (tears, pin holes, and cracks) accelerate the degradation. For Pt-based catalysts, particle dissolving/redeposits (known as Ostwald ripening), agglomeration, and redistribution contribute to a growth in the size of Pt particles [42,43]. At the same time, migration and microstructure break-ups due to carbon support corrosion gradually increase and lead to Pt loss. In the late age of degradation, obvious performance degradation is observed compared to the BOL, e.g. $\sim 10\%$ rated power loss. Interactive intensification and positive feedback mechanism appear which lead to a quick and unpredictable failure of stack. Pin holes, tears, and cracks appear and spread in the membranes, leading to gas penetration. In the Pt/C-based catalyst layer, a large amount of

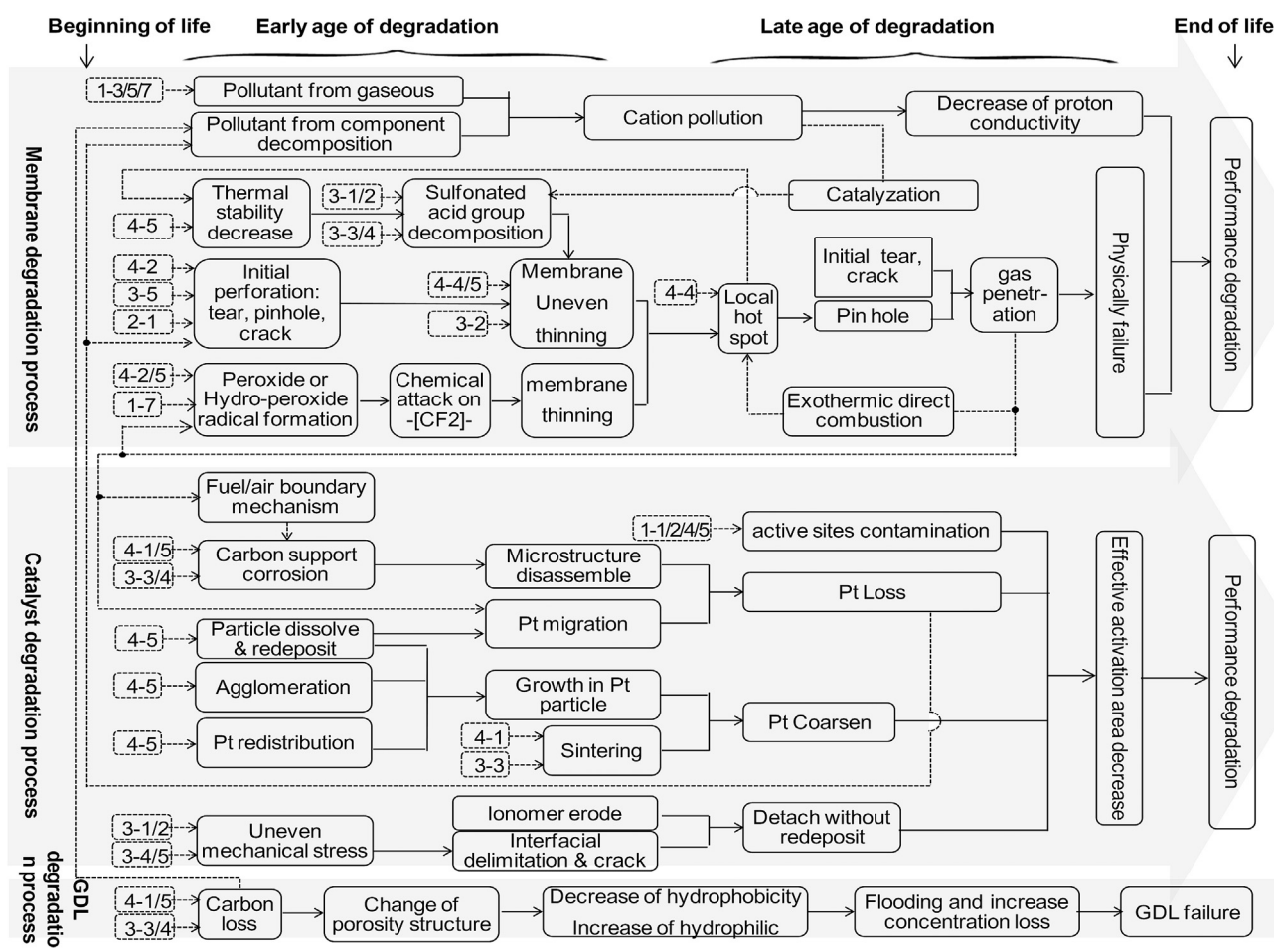


Fig. 2 – Cause and effect chain of the degradation of an automotive PEMFC.

Pt loss, particle growth, and crack and detachment decrease the effective activation area, resulting in an incremental loss of activation [44]. The early age of degradation process, to a large extent, determines the lifetime of the stack. So in this paper we only consider the direct intensification mechanism, and a linear voltage degradation rate assumption is used.

Key factors for the degradation of fuel cell voltage

The key factors that influence the long-term performance and durability are shown in Table 1.

The following factors and parameters are influenced by the load profile:

- 1) Corrosion of the electro-catalyst layers is a fundamental mechanism that strongly influences performance in the long run [37]. Cell potential cycling (e.g. from 0.1 V to 1.0 V or higher) is the most important factor contributing to platinum agglomeration and/or oxidation and hence to a decrease in the electrochemically active surface area [42].
- 2) Potential cycling (particularly on a high level and under high constant voltages) and lower relative humidity of the reactant gases aggravate the loss of carbon, leading to the corrosion of the Pt/C catalyst and the gas diffusion media [37].

- 3) High current density and overcurrent tend to cause local hot spots on unevenly-thinned membranes, which facilitates the chemical attack on the polymer end groups, thus aggregating membrane degradation. Hot spots can also affect the Pt-catalyst because high temperatures accelerate kinetics and therefore aid Pt-sintering [37].

Table 1 – Key factors affecting the individual components of the PEMFC.

Component	Key factors affecting degradation
Catalyst layer	High potential, load cycling, temperature, humidity, contaminants
Carbon support	High potential, high temperature, high water content, low PH, oxygen-rich environment,
Membrane	Open circuit voltage, load cycling, high temperature and low Humidity, freeze/thaw cycling
Gas diffusion layer	High potential, potential cycling, high temperature, oxygen
Bipolar plate	Cell reversal
Other components	Thermal cycling, humidification cycling

- 4) Fuel cell flooding and membrane dehydration are of vital importance to maintain the power density of PEM fuel cells in the long run. An excess of water accelerates the corrosion of the electrodes, the catalyst layers, the gas diffusion media, and the membrane [37]. Cathode flooding is usually found at high current densities because i) more water is created by the oxygen reduction reaction, and ii) a higher electro-osmosis flux occurs when an electric field is applied across the membrane. Anode flooding occurs more commonly at low current densities [37]. In experiments conducted by Ge and Wang [45], liquid water was only detected at the anode when the cell was operated at low current densities (0.2 Acm^{-2}). Membrane dehydration is not directly caused by the current but by insufficient humidification, water evaporation, and the electro-osmosis effect [46].
- 5) Extra-high current density and poor fuel supply management can cause gas starvation, which accelerates the corrosion of carbon components through a series of reactions. Poor water management due to flooding and poor heat management during sub-zero temperatures can cause local gas starvation [47].
- 6) Thermal and humidification cycling during operation in an automotive PEM fuel cell stack are usually caused by load cycling, although thermal management and water management subsystems can buffer the temperature and hydration state against load disturbances. Furthermore, thermal and water management effects are interactive, as in heat pipes, which can cause complex results with regard to performance degradation [48].

The general conclusions based on scientific observation and literature data are 1) a constant current density (galvanostatic) operation within a proper range of cell voltage is most favorable for the lifetime of the fuel cell; 2) a fluctuation in current density enhances the stack's degradation. The higher absolute rising/decreasing current density rate ($\text{Acm}^{-2} \text{ s}^{-1}$) it loaded, the severer voltage degradation is. 3) Low current density operation induces extra-high cell voltage. The higher voltage operates, the heavier the effect on the degradation is. An open circuit voltage is an extreme case. 4) High current density operation induces flooding, starvation, uneven current distribution, and hot spots. The higher the current density, the heavier the effect on the degradation is. A

qualitative description of the severities of the key factors is shown in Table 2.

A qualitative description of the effects of a load profile on degradation is necessary for the numeric prediction of the lifetime. This study aims to determine a method to describe numerically the relationship between voltage degradation rate and load profiles.

The characteristic value of the load profile

The characteristic value of the load profile is proposed as a key parameter to predict the voltage degradation rate. This value has two parts: one is based on the load current and the other on the load voltage. For normalization purpose, the voltage and current density of a single cell were used. Thus, the load current part was defined on a series of sampled current densities $i(n)$, while the load voltage part was defined on a series of sampled cell potentials $v(n)$.

1) Characteristic value of the load current

In order to measure the current magnitude and the cycling speed level, a spectrum of load current was used. Fig. 3 shows how to calculate the characteristic value of the load current.

First, a discrete Fourier transform was performed for the sampled current density series $i(n)$, and the frequency domain spectrum $F(\omega)$ was obtained:

$$F(\omega) = \text{DFT}[i(n)] \quad (1)$$

Because the transient response of the PEMFC is limited by the mass transfer process and many systems require a limited transient response during testing and application [58,59], the current frequency has an upper boundary $\omega_{\max}$.

Second, a weight function was applied. The weight function, $w_{\text{cur}}(\omega)$, was used to formulate synthetically the severities of the current profile. Based on the descriptions in A general review: cause and effect chain of automotive PEMFC degradation section, the weight function increases with increasing frequency. For simplification, a linear function is suggested:

$$w_{\text{cur}}(\omega) = k_1\omega + 1 \quad (2)$$

Third, the characteristic value of the load current (or its discrete formation) is calculated as:

Table 2 – Severities of the key factors affecting a PEMFC's degradation.

Categories	Factors	Membrane	Pt catalyst	Carbon support	GDL	Bipolar plate
Load profile feature (in current)	Constant current	—	—	—	—	—
	Current cycling	↓	↓ ⁺	↓	↓ ⁺	↓
	Extra-low current density	—	↓ ⁺	—	—	—
	Extra-high current density	↓ ⁺⁺	↓ ⁺⁺	↓	↓ ⁺⁺	—
Load profile feature (in voltage)	Galvanostatic operation	—	—	—	—	—
	Potential cycling	—	↓ ⁺⁺	↓ ⁺	↓ ⁺⁺	↓
	Open circuit voltage	—	↓ ⁺⁺	↓ ⁺⁺	↓ ⁺	—
	Extra-low voltage operation	↓ ⁺	—	—	—	—

“↓”: The factor contributes to the degradation of the corresponding component.

“—”: No effect.

“+” and “++”: Large degree of severity.

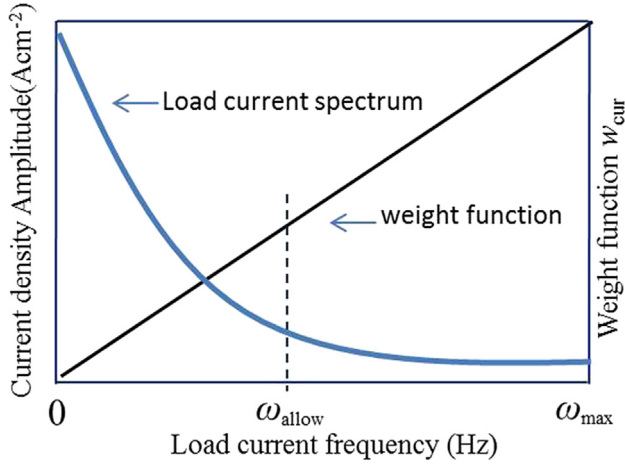


Fig. 3 – Load current spectrum and its weight function.

$$m_{cur} = \begin{cases} \frac{1}{T} \int_0^{\omega_{max}} F(\omega) w_{cur}(\omega) d\omega + 1 \\ \frac{1}{N} \sum_{n=1}^N F(\omega_n) w_{cur}(\omega_n) + 1 \end{cases} \quad (3)$$

2) The characteristic value of the load voltage

In order to measure the voltage effect, a voltage histogram of the load voltage was used. Fig. 4 shows how to calculate the characteristic value of the load voltage.

First, the load voltage frequency density, $H(v)$, was determined based on the voltage-time profile, with $[v_L, v_U]$ as the most suitable voltage range with regard to durability.

Second, a weight function was applied. The weight function $w_{vol}(v)$ represents the severities according to the details described in [A general review: cause and effect chain of automotive PEMFC degradation](#) section: 1) the lowest value during the suitable range of operation $[v_L, v_U]$; 2) a higher value in low current and overcurrent; 3) an extremely high value at open circuit voltage. The suggested weight function is:

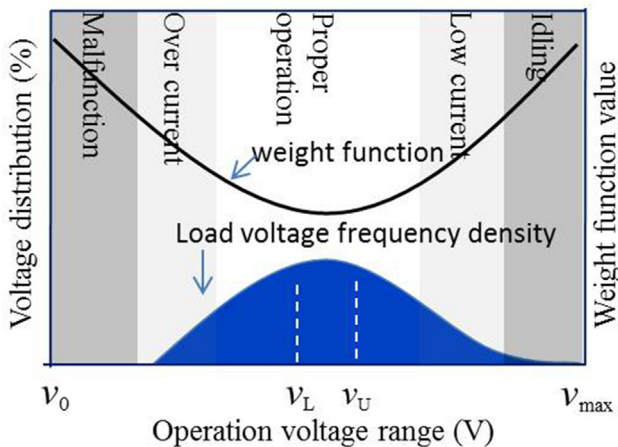


Fig. 4 – Load voltage histogram and its weight function.

$$w_{vol}(v) = \begin{cases} [k_2(v - v_L)]^2 & +1; v < v_L \\ 1; & v_L < v < v_U \\ [k_2(v - v_U)]^2 & +1; v > v_U \end{cases} \quad (4)$$

Thus, the characteristic value of the load voltage is:

$$m_{vol} = \begin{cases} \frac{1}{v_{max} - v_0} \int_{v_0}^{v_{max}} H(v) w_{vol}(v) dv \\ \frac{1}{N} \sum_{n=1}^N H(v_n) w_{vol}(v) \end{cases} \quad (5)$$

For $\int_{v_0}^{v_{max}} H(v) dv = 1$, m_{vol} is then:

$$w_{vol}(v_{suit}) \leq m_{vol} \leq w_{vol}(v_{max}) \quad (6)$$

3) The characteristic value

The product of m_{cur} and m_{vol} was defined as the load profile's characteristic value:

$$\varphi = m_{cur} \times m_{vol} \quad (7)$$

In Equation (1)–(7), the constant parameters are determined as follows:

- 1) The upper boundary of the frequency ω_{max} is physically determined by its mass transfer rate, and in a digital system, it is typically limited by the sampling period.
- 2) The upper boundary of the voltage v_{max} is determined by the thermal dynamic potential, which is in the range of 1.118–1.229 V. Although an even higher potential was observed during startup, it could be explained by other phenomenon and was relatively transient [60]. Thus, a sample data range between 0 and 1.229 V for the load voltage histogram was recommended.
- 3) A voltage range of [0.6 0.8] has been suggested as the most suitable operating range. Pourbaix diagrams show that Pt is unstable at a potential range of 1.0–1.2 V. The work by Borup et al. [49] has shown that the loss of the electrical chemistry surface area was approximately 5–10 times higher under potential cycling from 0.1 to 1.0 V than under potential cycling from 0.1 to 0.75 V. For reasons of current density, a low limit of operating voltage above 0.5 V has been recommended. Thus, we used [0.6 0.8] for $[v_L, v_U]$.

Fuel cell lifetime modeling with load profile

Lifetime definition using voltage degradation rate

A fuel cell's lifetime is defined as:

$$T_{cell} = \frac{0.1 \times V_{rated}}{\dot{v}(\varphi)} \quad (8)$$

where T_{cell} is the nominal lifetime, $0.1 \times V_{rated}$ is the allowable degradation voltage and $\dot{v}(\varphi)$ is the average voltage degradation rate.

Fuel cell voltage model with load profile terms

1) Fuel cell steady state model

The hydrogen fuel cell steady state voltage model is:

$$V_{\text{cell}} = E_{\text{Nernst}} - \eta_{\text{act}} - \eta_{\text{ohmic}} - \eta_{\text{conc}} \quad (9)$$

In Equation (9), the thermal dynamic potential, E_{Nernst} , is determined only by a reactant and the product's status, which can be assumed as constant if the stack operates under constant conditions. However, the voltage losses associated with the anode/cathode activation, proton/electron transfer, and mass transfer, namely η_{act} , η_{ohmic} , and η_{conc} , change during long-term operation.

The activation loss depends on the activation energy of the redox. The most commonly used electrode materials available today in the automotive industry are platinized or carbon particle-supported platinum (and in this paper, the model is for a Pt/C based fuel cell). The total activation overvoltage can be represented as the Tafel-equation [50,51]:

$$\eta_{\text{act}} = a + b \ln(i) \quad (10)$$

where a and b are coefficients.

The ohmic loss results from the resistance to the electron transfer in the bipolar plate and the GDL, as well as the resistance to proton transfer in the solid polymer membrane. The ohmic loss is formulated by Ohm's law. The total inner resistance R_{inner} can be obtained by evaluation and measurement of the parameters [52]:

$$\eta_{\text{ohmic}} = R_{\text{inner}} \cdot i = (R_{\text{elec}} + R_{\text{proton}}) \cdot i \quad (11)$$

The concentration loss results from the mass transfer and is a function of the limiting current density:

$$\eta_{\text{conc}} = \frac{RT}{nF} \ln \left(1 - \frac{i}{i_L} \right) \quad (12)$$

The limiting current density i_L depends mainly on the mass transfer coefficient, namely the diffusion property of the flow field and the GDL. Since the actual limiting current density is reduced by the current loss, the apparent limiting current density is used in substitution and can be obtained experimentally [53].

The limiting current density increased greatly as the PEMFC's performance improved. For example, in the 1990's it was measured at 0.9 [54], 0.81, and 2.5 Acm^{-2} [55]; the early 2000's, Santarelli et al. [52] reported an i_L of 9 Acm^{-2} at the cathode side versus 43 Acm^{-2} at the anode side.

2) Introduction of the aging and load profile terms

From Equation (10), a simplified long-term activation loss is:

$$\eta_{\text{act}} = a_0 + \delta_a(\varphi) \cdot t + b_0 \ln(i) \quad (13)$$

where $\delta_a(\varphi)$ is the increasing rate of a , which is formulated as an empirical function of φ .

In Equation (11), the electrical R_{elec} and the ionic resistance R_{iron} increased with operating time. The electrical impedance

spectrum experiment carried out by Diard et al. [23] indicated that the membrane resistance of a single cell contributed ~40% toward its overall ohmic resistance. Amphlett et al. [57] introduced another analytical expression of the ionic resistance, which was deduced from the analysis of literature data related to different cells of a Nafion 117 membrane. Within the formula, water content was treated as a parameter that changed at a constant rate over time [22]. In addition, the electrical resistance increases because 1) corrosion of the plates leads to contact resistance; and 2) mechanical stress cycling results in delamination of the membrane electrode assembly.

To simplify the expression, all the factors are merged, and as a result, the ohmic loss is:

$$\eta_{\text{ohmic}} = [R_0 + \delta_R(\varphi) \cdot t] \cdot i \quad (14)$$

The increasing rate of inner resistance $\delta_R(\varphi)$ is also an empirical function of φ . Let the limiting current change with time, and the concentration loss is:

$$\eta_{\text{conc}} = \frac{RT}{nF} \ln \left(1 - \frac{i}{i_L(t)} \right); i_L(t) = i_{L0} + \delta_L(\varphi) \cdot t \quad (15)$$

where $\delta_L(\varphi)$ is the increasing rate of the limiting current density i_L and is formulated as an empirical function of φ .

By replacing the time-varied items of activation loss, the ohmic loss, and the concentration loss with Equations 13 and 14, And (15), the voltage degradation model is:

$$V_{\text{cell}} = E_{\text{Nernst}} - \left[a_0 + \delta_a(\varphi) \cdot t + b_0 \ln(i) + (R_0 + \delta_R(\varphi) \cdot t) \cdot i + \frac{RT}{nF} \ln \left(1 - \frac{i}{i_{L0} + \delta_L(\varphi) \cdot t} \right) \right] \quad (16)$$

Voltage degradation rate

The concentration loss can be omitted for the load current density and is far smaller than i_L at the rated power. Thus, the voltage degradation rate is obtained by differentiating Equation (16) with t :

$$\dot{V}(\varphi) = - \left. \frac{dV_{\text{cell}}}{dt} \right|_i = \delta_a(\varphi) + i \delta_R(\varphi) \quad (17)$$

With the load profile characteristic value φ as the only variable in the voltage degradation rate formula, it is easy to determine a proper set of coefficients, namely k_1 , k_2 in Equations (3) and (5), to fit Equation (17).

Modeling result and discussion

It is rather painstaking to carry out durability experiments with all possible load profiles because it is time-consuming and expensive. Hence, in this paper, the following method was used to validate the model. First, publically reported voltage degradation data under different load profiles were collected, and the coefficients in the model were fitted based on these data. Second, a durability experiment was carried out under specific load profiles and was used as a special case to validate the model.

Experiment data collection

Table 3 provides a selected collection of literature data on voltage degradation rates under different load profiles for a perfluorinated ionomer membrane-based electrolyte and a Pt/C-based PEMFC.

The following notes apply to Table 3:

- 1) In the first, second, and third column, numbers of the study cases, study times, and group names are listed chronologically;
- 2) In the fourth, fifth and sixth columns, the experiment's duration and load profile features are listed and the voltage degradation rate at the specific current density is marked;
- 3) In the last column, the characteristic values of the load profile are calculated. First, the load current spectrums were determined based on Equation (1). For the galvanostatic load, $F(0) = i$ and $F(\omega, \omega \neq 0) = 0$; $H(v \neq v_{\text{static}}) = 0$ and $H(v_{\text{static}}) = 1$. For the other case, the load profile was defined in the paper or based on the referenced testing specifications. Second, the characteristic values from the current part and the voltage part were calculated based on Equations (2), (5) and (7). The values of No. 1–5, 9, 11, and 12 were nearly identical except in the case of No. 8 due to the higher voltage. Some of the load profiles were not achieved or were unclear and are thus marked with “N/A”;
- 4) The results marked with “*” (No. 15 and 21) were carried out by the authors and their group. The load profile had an average cycle time of 800 s and an average current density, an acceleration rate, and a deceleration rate of 0.140 mA cm^{-2} , $1.32 \text{ mAcm}^{-2} \text{ s}^{-1}$, and $-1.2 \text{ mAcm}^{-2} \text{ s}^{-1}$, respectively. The maximum load acceleration and deceleration were $4 \text{ mAcm}^{-2} \text{ s}^{-1}$ and $-36.8 \text{ mAcm}^{-2} \text{ s}^{-1}$, respectively.

Fit of coefficients

In Table 3, the reported degradation rates occurred at different current densities of i in the range of $0.02\text{--}0.7 \text{ Acm}^{-2}$. To compare the degradation rates from different experimental results, a value of 0.5 Acm^{-2} was chosen because it represents a typical density. Most of the results in Table 3 were approximately 0.5 Acm^{-2} , and a correction was applied when the value was outside of it. A linear function was used to fit the relationship between the degradation rate and the load's characteristic value. The voltage degradation rate $\dot{v}(\varphi)$ was then:

$$\dot{v}_{0.5} = \left. \frac{dv_{\text{cell}}}{dt} \right|_{0.5} = \alpha_{0.5} \cdot \varphi(k_1, k_2) + \beta_{0.5} \quad (18)$$

Here, $\alpha_{0.5}$ and $\beta_{0.5}$ are the linear fitting coefficients when $i = 0.5 \text{ Acm}^{-2}$. Let the coefficients k_1 and k_2 be constant numbers (e.g. 25 and 10) and the linear relationship between voltage degradation rate and φ was obtained as:

$$\dot{v}_{0.5} = \delta_a(\varphi) + 0.5\delta_R(\varphi) = 40\varphi - 46 \Big|_{\varphi(25,10)} \quad (19)$$

In Fig. 5, the result marked with “*” was carried out by the authors, and it indicates that the model is able to predict the voltage degradation level.

Let k_1 and k_2 be 1–25 and 1–10, and fit the relationship between $\varphi(k_1, k_2)$ and the voltage degradation rate. The goodness of fit is as shown in Fig. 6.

It is found that the coefficients k_1 and k_2 were not suitable for the fitting results because the goodness-of-fit was always ≥ 0.90 when $k_1 > 10$ and $k_2 > 5$. The coefficients k_1 and k_2 could be variable, and we recommend $k_1 > 25$ and $k_2 > 5$ based on our results.

The weight function of Equations (3) and (5) proved suitable to weight the effects of load current and voltage on the degradation of the PEMFC.

Table 3 – Voltage degradation rates under different load profiles.

No.	Year	Researchers	Time(h)	Load profile/cycling feature	$\dot{v}(\varphi)(\mu\text{Vh}^{-1})$	Characteristic value φ
1	1997	Ralph [61]	1000	Galvanostatic, 0.646 Acm^{-2}	4.0	1
2	2000	St. Pierre et al. [62]	5000	Galvanostatic	1.0	1
3	2002	St. Pierre, & Jia [63]	11000	Galvanostatic,	2.0	1
4	2002	Fowler et al. [22]	1350	Galvanostatic,	11	1
5	2004	Endoh et al. [64]	4000	Galvanostatic	2.0	1
6		Id.	160	Accelerated OCV test	2000	N/A
7		Id.	160	Normal OCV test	200	N/A
8	2005	Ferreira et al. [65]	2000	Galvano static, $0.95\text{V}@0.0 \text{ Acm}^{-2}$	50	$(0.0225k_2^2 + 1)$
9		Id.	2000	Galvano static, $0.75\text{V}@0.2 \text{ Acm}^{-2}$	25	1
10		Id.	111	Voltage cycles $0.6\text{--}1.0\text{V}@20 \text{ mV/s}$	300	$(0.1434k_1 + 1)(0.0066k_2^2 + 1)$
11	2006	Cleghorn et al. [66]	26300	Galvano static, 0.30 Acm^{-2}	6.0	1
12	2007	Wahdame et al. [67]	1000	Galvano static, 0.5 Acm^{-2} Gore	30	1
13		Id.	700	Vehicle mission profile. Gore	75	N/A
14	2007	Owejan et al. [68]	–	20°C , start-stop cycling	212	N/A
15	2008*	Lin et al. [69]	370	Driving cycle protocol	270	$(0.1534k_1 + 1)(0.0206k_2^2 + 1)$
16	2009	Wang et al. [70]	2250	160 mAcm^{-2}	Unknown	N/A
17	2010	Franck-Lacaze et al. [71]	530	$120\text{h}@0.12 \text{ Acm}^{-2}$, $410\text{h}@0.54 \text{ Acm}^{-2}$	50	$(0.111k_1 + 1)(0.002k_2^2 + 1)$
18		Id.	500	$270\text{h}@0.02 \text{ Acm}^{-2}$, $230 \text{h}@0.54 \text{ Acm}^{-2}$	400	$(0.1347k_1 + 1)(0.011k_2^2 + 1)$
19	2010	Weng et al. [72]	150	Current cycling, $0.7\text{--}0.07 \text{ Acm}^{-2}$	200	$(0.1489k_1 + 1)(0.005k_2^2 + 1)$
20	2010	Hou et al. [73]	1500	Beijing City cycle, on road	45	N/A
21	2010*	Zhang et al. [74]	2000	Expo cycle, on road	60	$(0.0911k_1 + 1)(0.01k_2^2 + 1)$

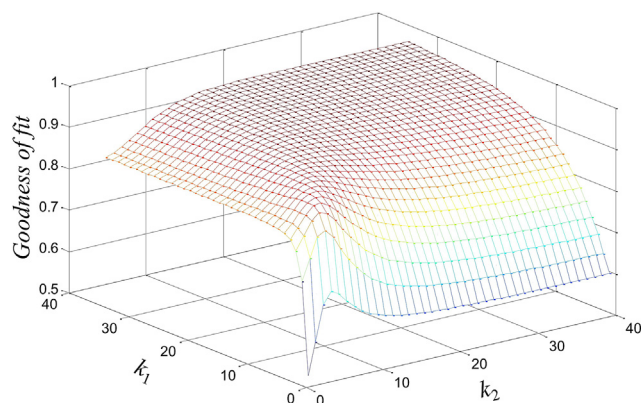


Fig. 5 – Linear relationship between voltage degradation rate and ϕ .

Discussion

This model was able to deliver a rapid evaluation of the PEMFC's voltage degradation level by using only the load profile, which can be obtained easily through numerical simulation during the design of the automotive powertrain. Thus, the model creates a close connection between the performance of the automotive powertrain and the PEMFC's durability.

However, there are still some disadvantages and limitations:

- 1) Data from different studies was used involving different types of PEM fuel cell stacks, which consisted of different materials and components. Our data reflect only the degradation level, but not the exact degradation rate. Hence, the prediction results from the model are only approximate.
- 2) The voltage degradation rates differed at different current densities, which mainly depended on their polarization performance. For lack of data, the parameters δ_a and δ_R could not be estimated, and thus the results are less accurate when predicting voltage degradation rates at current densities other than 0.5 Acm^{-2} . Multiscale modeling and simulation method could have deep insight in degradation process spanning scales from few atoms to the

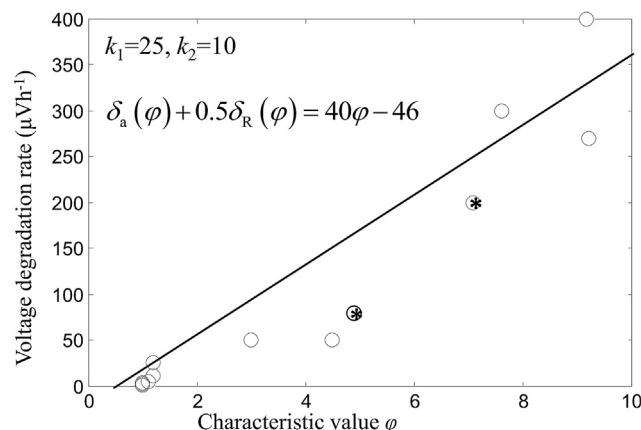


Fig. 6 – Goodness of fit with linear expression.

device level [75,76,77]. The method would link material degradation to macroscopic performance behavior, and distinguish δ_a and δ_R respectively.

Conclusions

The key factors affecting the performance degradation of a PEMFC were analyzed. The mechanisms of degradation were described in the cause and effect chain and the factors were attributed to load current and voltage.

The characteristic value of the load profile was proposed and this value was based on the features of the load current and load voltage. The current density spectrum and voltage histogram together with weight functions were used to determine the characteristic value, which allows for measuring and comparing different load profiles and for determining the effects on the PEMFC's degradation.

A voltage degradation model with both aging and load profiles was proposed. The model was based on the steady state electrochemical model. Aging terms and load profile terms were introduced into the activation loss, the ohmic loss, and the concentration loss respectively.

The degradation rate of the cell's voltage was a linear function of the load profile's characteristic value, which facilitates the prediction of the lifetime of the PEMFC.

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